DEPARTMENT OF WORKS & HIGHWAYS
MATERIAL SCIENCE & TECHNOLOGY BRANCH
Materials Testing Laboratory

☑ Port Moresby ☐ Mt. Hagen ☐ Lae ☐ Annex Lab.

Date:

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FIELD DENSITY (SAND REPLACEMENT METHOD) AS1289.5.3.1

Client : _____ Asset No.: _____

Project: _____ Job No: _____

Location: _____ Material: Fill / Subgrade / Sub base / Base / Others

MASS OF SAND TO FILL CONE AND TRAY

M1	Initial mass of sand & bottle	3000	3000	3000
M2	Final mass of sand and Bottle	875	870	880
M3	Mass of sand (fill cone and	2125	2130	2120
	Mean mass of sand to fill cone & tray		**2125**	

CALIBRATION OF FIELD DENSITY SAND

	Cylinder Volume (V)	2600
M4	Initial mass of sand and bottle	7,000.00
M5	Final mass of sand and bottle	
M6	Mass of sand to fill calibrating cylinder (g	
	Pouring density of sand (M6/ V)	

	Test No							
	Location of test / Chainage							
	Offset of Centrline (Left, Center-line, Right)		**(m)**					
	Depth Level		**(mm)**					
	Field Remarks:							
	SURFACE CORRECTION							
M7	Mass of sand &Bottle		**g**					
M8	Mass of sand residue & Bottle		**g**					
M8a	Mass of sand used to fill cone (M7 - M8)		**g**					
	VOLUME OF SOIL & ROCK							
M9	Mass of sand & Bootle		**g**					
M10	Mass of sand residue & Bottle		**g**					
M12	Mass of sand to fill excavated hole (M9 - M10) - (M8a)		**g**					
Vh	Volume of hole (M12)/(*P* sand)		**cm³**					
	MASS OF SOIL AND ROCK							
	Container Number							
Ms	Mass of excavated soil & container		**g**					
Mc	Mass of container		**g**					
M11	Mass of excavated soil Ms - Mc		**g**					
M14	Mass of excavated rock retained on 19mm sieve		**g**					
M15	Mass of excavated soil (passing 19mm seive).		**g**					
M16	Mass of dry rock		**g**					
M17	Mass of dry soil 100 x M15/(100+M35)		**ml**					
M18	Total dry mass (M16) + (M17)		**g**					
M19	Mass of surface dry rock		**g**					
M20	Volume of water displaced		**ml**					
M21	Apparent Density M19/M20		**t / m³**					
M22	Volume of rock M14/M21		**cm³**					
M23	Percentage of rock 100 x M14/M11		**%**					
M24	Percentage of soil 100 - (M23)		**%**					
M25	Total wet density M11/Vh		**t / m³**					
M26	Total Dry density 100 x M25/(100+M35)		**t / m³**					
M27	**Maximum Dry Density**		**t / m³**					
	c = M24 / M27 + M23 / M21							
M28	Adjusted MDD 100/ c		**t / m³**					
M29	Adjusted OMC (M36) x (M24)/100		**%**					
M30	**Density Ratio** [(M26) / (M28)] 100							
	DETERMINATION OF MOISTURE CONTENT							
	Container Number							
M30	Mass of wet soil & container		**g**					
M31a	Mass of dry soil & container		**g**					
M31b	Check Mass of Dry soil & container		**g**					
M32	Mass of container		**g**					
M33	Mass of water M30 - M31		**g**					
M34	Mass of dry soil M31 - M32		**g**					
M35	Field moisture content 100 x M33/ M34		**%**					
M36	**Optimum Moisture Content**		**%**					
M37	Variation from OMC (dry/wet) (M36) - (M35)		**%**					

Tested by: _____ / / **Checked By** _____ / /

Authorised by: AS (Material Science & Technology Branch)

FORM No. S-5.3.1

Revised Worksheet: 08/04/2022 lhb